

AuraEnroll AI

Aura Integrated University Enrollment Portal

Intelligent Conversational Enrollment System with Multi-Role AI Ecosystem

Document Type	Project Proposal — Academic Submission
System Name	AuraEnroll AI — Enrollment Portal
Development Team	Aura Integrated Software Solutions
Institution	Aura Integrated University (AIU)
Status	Phase 4 — Final Validation (In Progress)
Server	http://localhost:5000
Database	Supabase / PostgreSQL

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1. Executive Summary

AuraEnroll AI is a state-of-the-art academic enrollment portal engineered for Aura Integrated University (AIU). The system harnesses Autonomous Artificial Intelligence — branded as Aura AI and powered by the Groq Llama-3-8b model — to streamline the student onboarding process through natural conversational data intake. By eliminating traditional manual form submissions, AuraEnroll AI delivers a seamless, cinematic user experience while maintaining institutional-grade data integrity through a managed Supabase PostgreSQL backend.

The platform serves three distinct user roles — Student, Administrator, and Faculty — each supported by a dedicated AI-powered interface that automates routine tasks and provides intelligent real-time insights.

2. Project Objectives

- **Autonomous Data Extraction:** Implement Aura AI via Groq Llama-3 to accurately extract student entities — name, course, contact information, and address — from unstructured natural language chat.
- **Robust Data Integrity:** Ensure zero-loss data synchronization between the React frontend and the managed Supabase PostgreSQL database through Prisma ORM type-safe migrations.
- **Premium User Experience:** Deliver a high-fidelity, fully responsive interface featuring cinematic animations via Framer Motion and real-time form state feedback.
- **Backend Scalability:** Architect a high-performance Express.js and Prisma backend capable of handling concurrent enrollment operations with minimal latency.

3. Technology Stack

The following industry-standard technologies were selected to ensure stability, performance, and long-term maintainability of the AuraEnroll AI platform:

Layer	Technology	Role / Purpose
Frontend	React.js (Vite) + Tailwind CSS + Framer Motion	UI rendering, cinematic animations, state management
Backend	Node.js + Express.js	High-speed REST API gateway
ORM	Prisma 6.x	Type-safe schema migrations and queries
Database	Supabase — PostgreSQL (Managed)	Cloud-hosted persistent data storage (SSL)
AI Engine	Groq Cloud API — Llama-3-8b	Sub-second NLP, entity extraction, SQL generation

4. System Architecture

The architecture follows a layered, separation-of-concerns model. The three frontend portals communicate exclusively via RESTful API with the Express.js server, which orchestrates calls to the Groq AI engine and the Prisma ORM layer connected to Supabase PostgreSQL.

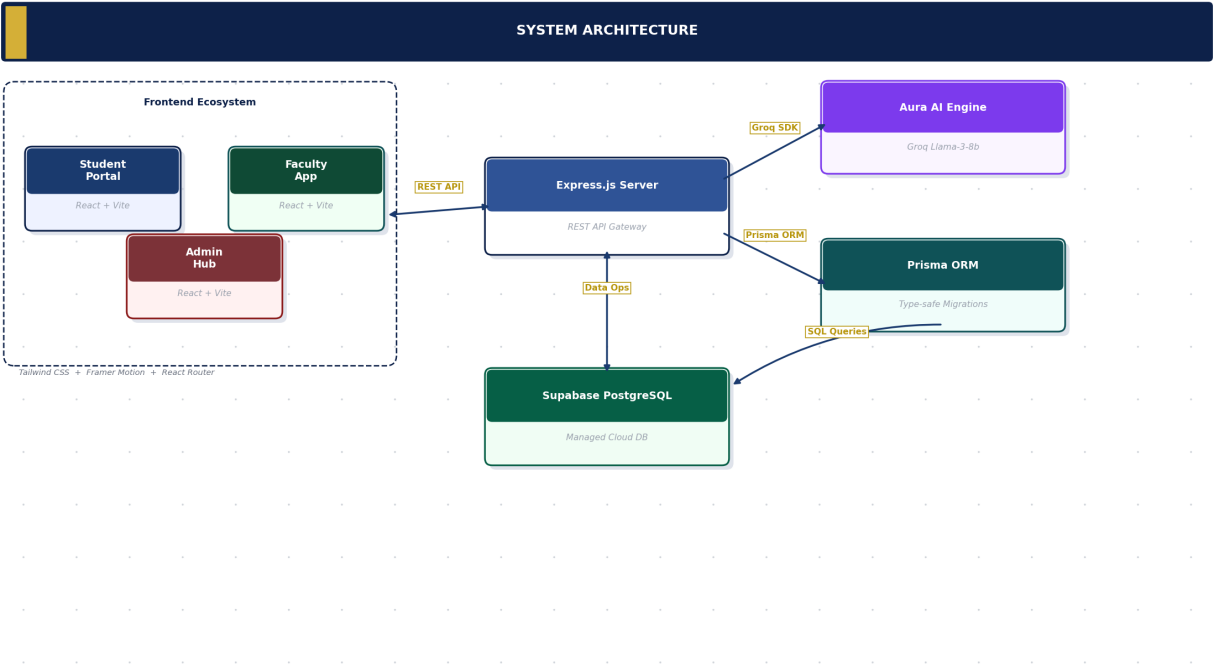


Figure 1. AuraEnroll AI — System Architecture Diagram

5. Multi-Role AI Ecosystem

AuraEnroll AI operates as a tri-role AI ecosystem, with each role powered by the Groq Llama-3-8b model but optimized for its specific institutional function:

Role	AI Module	Core Function
Student	Aura AI — Student Onboarding	Conversational enrollment; real-time entity extraction to auto-populate the registration
Administrator	Aura Admin — Intelligence Hub	Conversational analytics; autonomous SQL generation for instant institutional reporting
Faculty	Aura Faculty — Academic Assistant	AI-assisted class management; student profile organization, trend summarization, and

6. Process Flow Table

The following table documents the end-to-end process flow across all three user roles, mapping each step to its responsible actor, action, and resulting system output:

Step	Actor	Action Performed	System Output / Response
1	All Users	Navigate to the AuraEnroll Portal	Role selection screen is presented
2	Student	Engage in chat with Aura AI	Groq extracts entities; form auto-populated in real-time
3	Student	Review extracted form data	System presents 'Ready for Submission' confirmation state
4	Student	Confirm and submit application	Record committed to Supabase PostgreSQL database
5	Admin	Submit natural language analytics query	Aura generates underlying SQL from the prompt
6	Admin	View generated analytics report	Live charts rendered from real-time database records
7	Faculty	Command student profile organization	Schedules and profiles automatically updated in DB

7. Entity Relationship Diagram (ERD)

The ERD below illustrates the normalized database schema powering AuraEnroll AI. The **ENROLLMENT** entity serves as the system's primary data store. **CLASS_SECTION** records are managed by **FACULTY**, and enrollment records are associated with sections. **ADMIN** actions are audited through **SYSTEM_LOGS**. All primary keys (PK) use UUID format for distributed scalability. Foreign keys (FK) enforce referential integrity, and unique keys (UK) prevent duplicate entries on critical fields such as email and username.

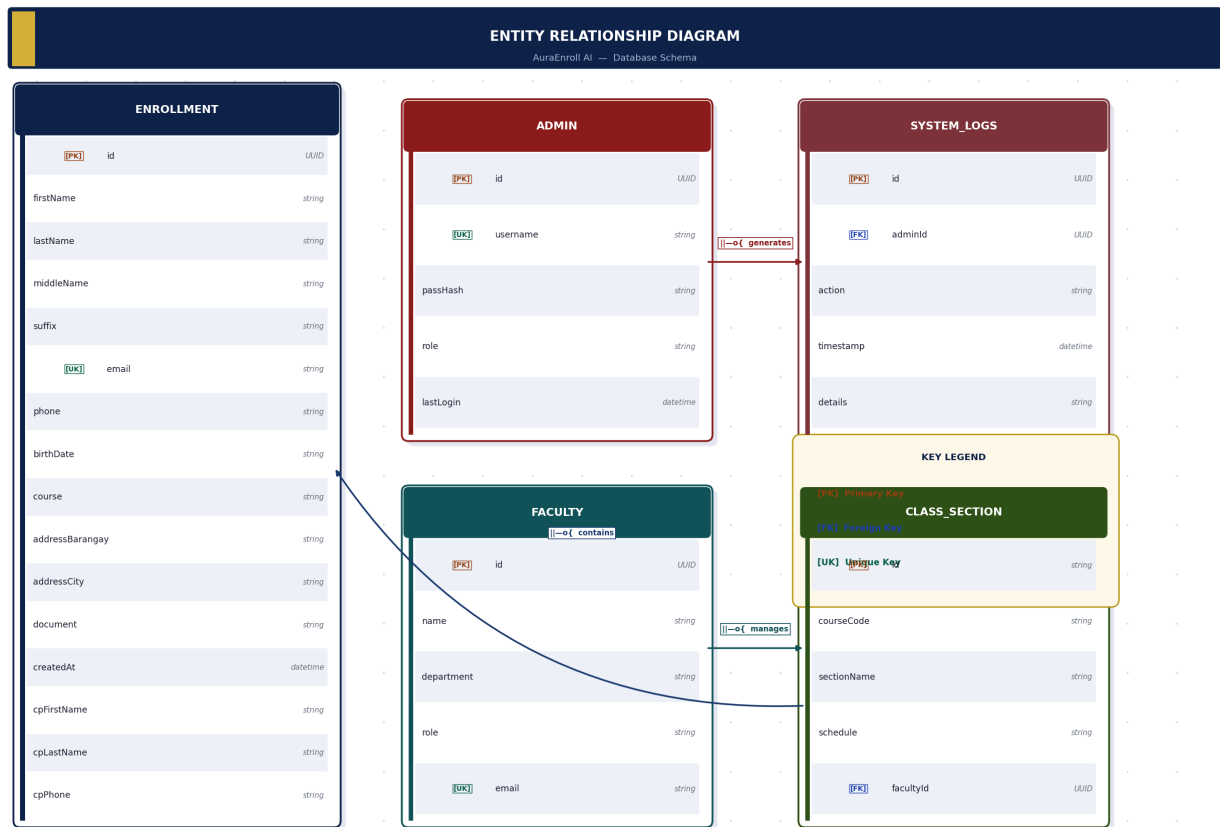


Figure 2. AuraEnroll AI — Entity Relationship Diagram (ERD)

8. Data Dictionary

The following data dictionary provides a detailed breakdown of each entity's attributes, data types, key classifications, and field descriptions:

Entity	Attribute	Type	Key	Description
ENROLLMENT	id	UUID	PK	Unique identifier for the enrollment record
	firstName	string		Student's first name
	lastName	string		Student's last name
	middleName	string		Student's middle name
	email	string	UK	Unique active email address

	phone	string		Student contact number
	birthDate	string		Student's date of birth
	course	string		Selected academic track or program
	addressBarangay	string		Student's registered barangay
	addressCity	string		Student's city or municipality
	document	string		Submitted document reference
	createdAt	datetime		Timestamp of application submission
	cpFirstName	string		Emergency contact's first name
	cpLastName	string		Emergency contact's last name
	cpPhone	string		Emergency contact's phone number
ADMIN	id	UUID	PK	Unique identifier for the administrator
	username	string	UK	Admin login credential (unique)
	passHash	string		Bcrypt-hashed password
	role	string		System permission level (e.g., SuperAdmin)
	lastLogin	datetime		Timestamp of most recent login session
FACULTY	id	UUID	PK	Unique identifier for the faculty member
	name	string		Faculty member's full name
	department	string		Assigned academic department
	role	string		Faculty designation or position
	email	string	UK	Faculty's unique institutional email
CLASS_SECTION	id	string	PK	Unique identifier for the class section
	courseCode	string		Associated academic course code
	sectionName	string		Human-readable section name
	schedule	string		Class schedule (day, time, room)
	facultyId	UUID	FK	Reference to the assigned Faculty member

9. System Process Flowchart

The flowchart below presents the complete user journey across all three system roles. Each column represents an independent workflow: the **Student Path** demonstrates AI-assisted conversational enrollment with a data confirmation loop; the **Admin Path** covers natural language analytics and chart generation; and the **Faculty Path** illustrates AI-driven class and profile management.

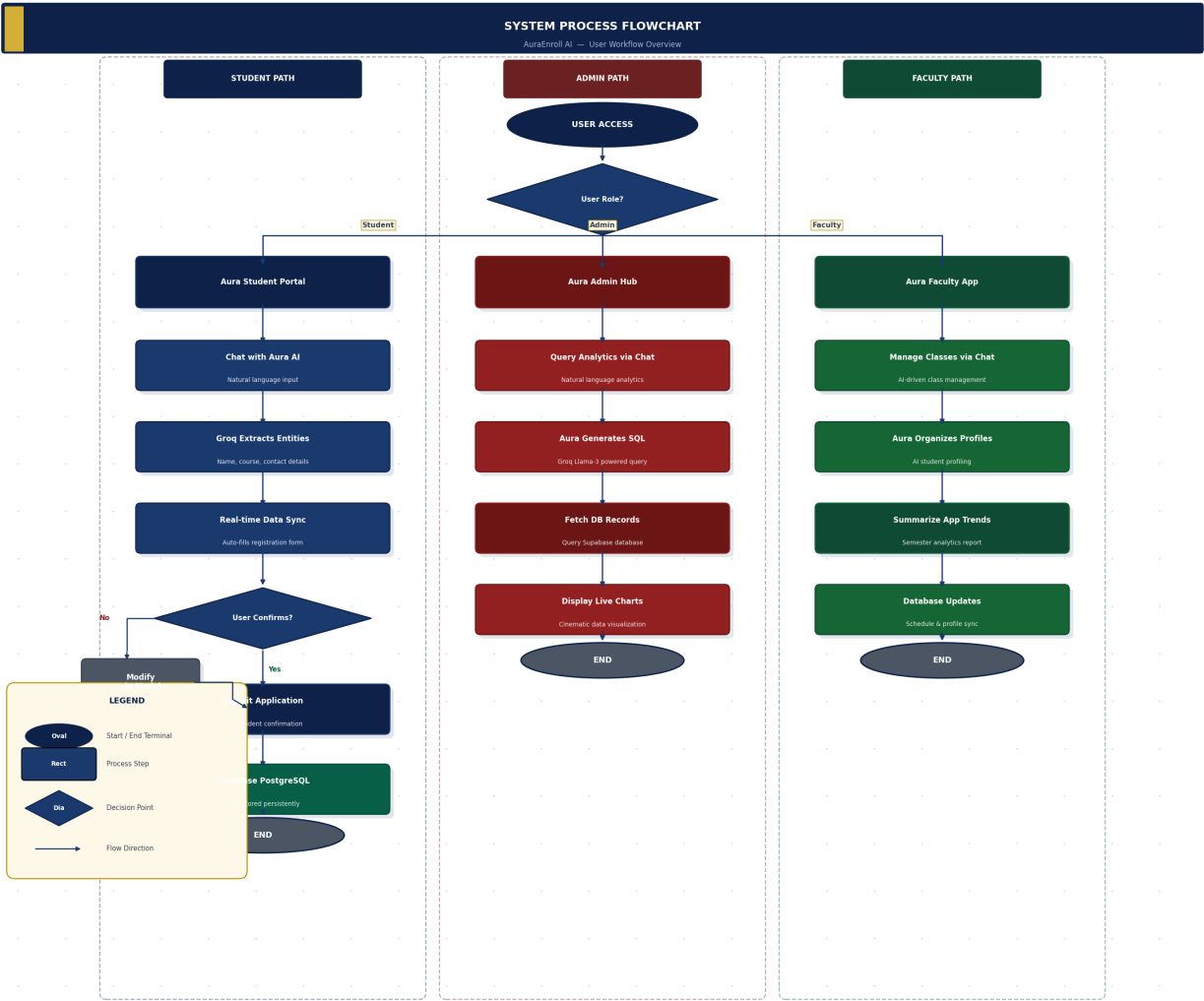


Figure 3. AuraEnroll AI — System Process Flowchart

10. Implementation Roadmap

Development follows a four-phase incremental delivery model, ensuring each layer is validated before proceeding to the next:

Phase	Milestone	Deliverables	Status
1	UI Foundation	Cinematic landing page, role-based routing, registration portal scaffolding	Completed
2	AI Brain Integration	Grok API integration, Aura AI conversational logic, entity extraction logic	Completed
3	Data Persistence	Prisma schema push, Supabase connection, full CRUD synchronization	Completed

4	Final Validation	End-to-end testing, performance benchmarking, production deployment	In Progress
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Institutional Server: http://localhost:5000 | **Database:** Supabase / PostgreSQL | **AI Engine:** Groq Cloud — Llama-3-8b